

Reshma Katharin Biju

(716) 907-5082 | reshma2001biju@gmail.com | www.linkedin.com/in/reshma-katharin-biju

EDUCATION

University at Buffalo, State University of New York — Buffalo, NY

Aug 2024 – May 2026

M.S. Biomedical Engineering (STEM), GPA: 3.87/4.00

Maharashtra Institute of Technology — India

Aug 2019 – May 2023

B.Tech Biomedical Engineering, GPA: 3.44/4.00

SUMMARY

Biomedical Engineering graduate researcher with experience developing machine learning and signal processing pipelines for electrophysiological data analysis, including EEG-to-EMG predictive modeling using NLARX and neural-network-based approaches. Skilled in MATLAB and Python for biomedical data preprocessing, feature engineering, visualization, and time-series analysis across heterogeneous neurophysiological datasets. Experienced in building reproducible computational workflows and AI-assisted modeling systems for biomedical research, with ongoing first-author manuscript preparation focused on EMG-based classification and task performance assessment. Strong interest in computational neuroscience, representation learning, and AI-driven biomedical data integration.

EXPERIENCE

Graduate Researcher, COMAND Lab, University at Buffalo — Buffalo, NY

Aug 2024 – Present

- Built EEG-to-EMG modeling pipelines using NLARX and neural networks for multi-subject task performance assessment, achieving 89% classification accuracy.
- Developed biomedical signal preprocessing, feature extraction, visualization, and model evaluation workflows in MATLAB and Python for quantitative electrophysiological analysis.
- Applied AI-assisted workflows for ML pipeline development and preparing a first-author manuscript on EMG-based classification for neurorehabilitation applications.

Breast Cancer Detection Using DNN, University at Buffalo — Buffalo, NY

Aug 2024 – Dec 2024

- Developed a deep neural network (DNN)-based biomedical classification model using structured preprocessing, validation, and performance optimization workflows.
- Implemented machine learning and data analysis pipelines in Python for high-dimensional biomedical data processing and predictive modeling.
- Achieved 91% AUROC and 83% precision through iterative model evaluation, quantitative analysis, and validation strategies.
- Built visualization workflows to support interpretable diagnostic prediction and reproducible computational analysis.

Biomedical Engineer Intern, Enduro I Pvt. Ltd. (R&D Prosthetics) — Kerala, India

Nov 2021 – Jan 2022

- Conducted user-centered analysis of upper-limb prosthetic systems by integrating biomechanical assessment data and usability feedback into design evaluation workflows.
- Evaluated myoelectric and body-powered prosthetic devices to identify functional limitations and opportunities for system-level optimization.
- Collaborated within an R&D environment to support iterative development of assistive biomedical technologies and improve device usability.

LEADERSHIP

President, BME Graduate Student Association — Buffalo, NY

Aug 2025 – Present

- Launched mentorship program (7 mentors) and Media Team to streamline communications and program documentation.
- Managed \$2,746 budget across 6+ initiatives, allocating \$874.25 in event expenditures with full financial accountability.
- Coordinated major events (Johnson & Johnson Career Talk, IGSA/Yalda Night) for 25–100+ students while handling high-pressure student needs and resource coordination.
- Led BME GSA to win the "Student Organization Collaboration of the Year" Award — a first in the organization's history.

PUBLICATIONS

Reshma Katharin Biju, et al.

May 2026 – Present

- EMG-Based Classification Using Nonlinear Autoregressive Exogenous (NLARX) Modelling for Task Performance Assessment — Manuscript in preparation for publication.

TECHNICAL SKILLS

Programming & AI/ML: MATLAB, Python, NumPy, Pandas, Scikit-learn, DNN, NLARX modeling, AI-assisted coding workflows

Signal Processing & Computational Analysis: EEG/EMG signal processing, electrophysiology data analysis, time-series modeling, biomedical signal preprocessing, model evaluation (RMSE, MAE, AUROC, CAM)

Medical & Laboratory Systems Delsys Trigno EMG, EEG systems, LabChart, Magstim TMS, NeuroConn tDCS, MRI-guided neuronavigation, IRB documentation, CITI Human Subjects Certification